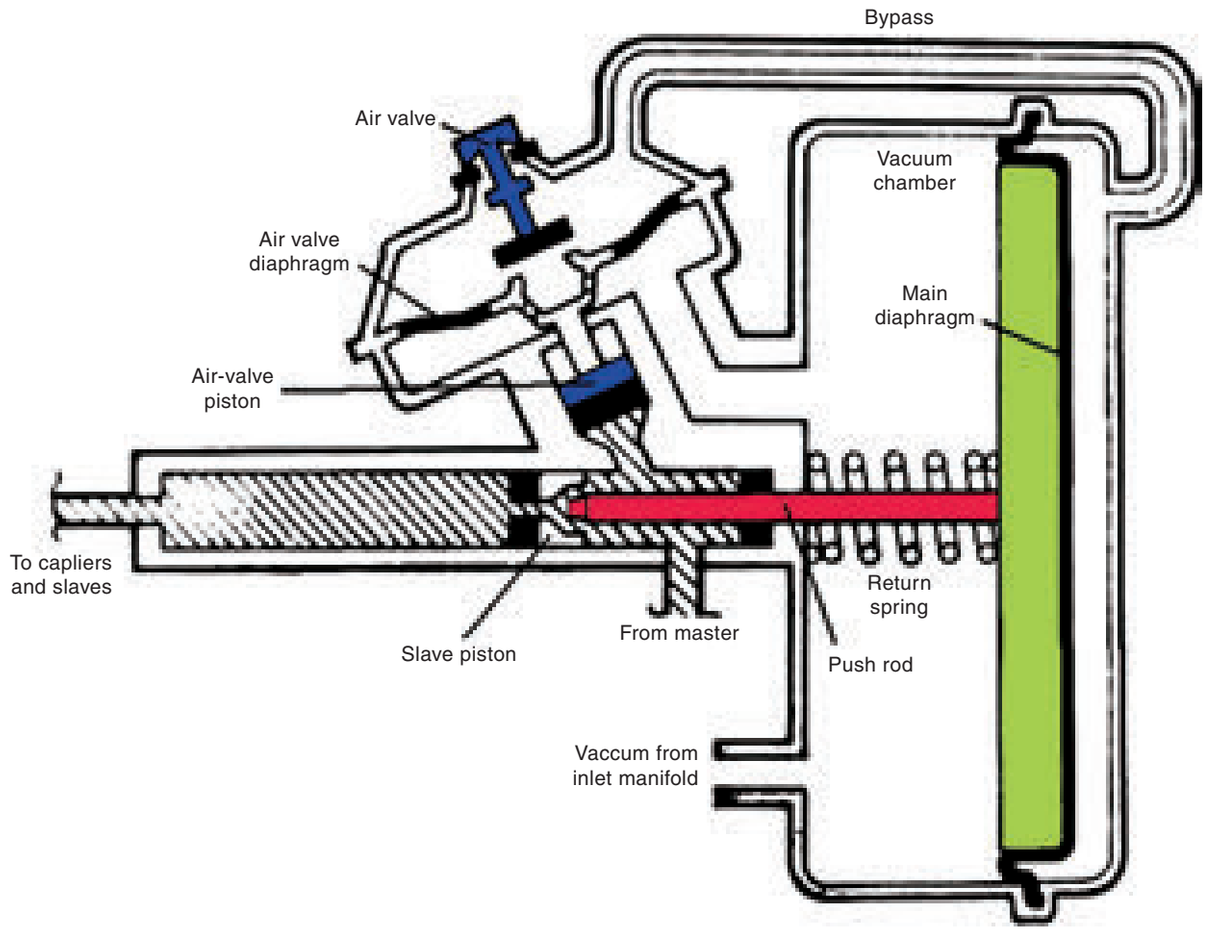


Figure 6.4.3.1 (a) Schematic of the Air assisted servo Brake system

Vacuum servo brake

The schematic view of the vacuum servo brake is shown in the figure. This type of brake system utilizes the vacuum created at the inlet manifold of engine and increases the braking force. The vacuum reservoir is connected to the inlet manifold of the engine with a non-return valve. Moreover, vacuum reservoir is connected on both sides of pistons inside the servo cylinder. In this way the vacuum reservoir is attached to the right side of the piston in the Servo cylinder and the left side is attached with the control unit and the schematic of vacuum assisted servo brake can be seen in Figure 6.4.3.2.

There is a piston in the control unit that has two valves attached to it. These valves are arranged in a compressed state with the help of two springs. The valve at the top of the control unit controls the atmospheric pressure controls the connections in the left side of the piston in the servo cylinder. Similarly, valve at bottom of the control unit controls the connections between the vacuum reservoir which is located in the right side of the control unit and servo cylinder. The piston inside the control unit is actuated by master cylinder. In the control unit valve at the top is in closed position and bottom will be in opened position when there is no any force applied to the brake pedal. Thus, the two sides of the piston inside servo cylinder are equally contacted with the vacuum in the inlet manifold.

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